

# HydroMesh

Street-level flood intelligence for safer cities

Community reports + climate data → faster local action

Output Outlaws · University of Birmingham · SDG 11

[hydromesh.vercel.app](https://hydromesh.vercel.app) · [github.com/theSaksham02/Hydromesh](https://github.com/theSaksham02/Hydromesh)



## Flood report

Knee-deep water  
Main St · Just now

**Submit report**

[Live map](#) · [SOS](#)

# When floods change by the minute, broad alerts are not enough.

## CURRENT SITUATION



Broad warnings, no street-level data



Fragmented reports from residents



Responders lack a unified operational view

## HUMAN CONSEQUENCE

- 1 Unsafe travel decisions
- 2 Delayed emergency response
- 3 Greater exposure for vulnerable residents

# Communities need local information before they can take local action.



## **Flood reports are fragmented**

Residents, responders, and agencies use different channels



## **Conditions vary street by street**

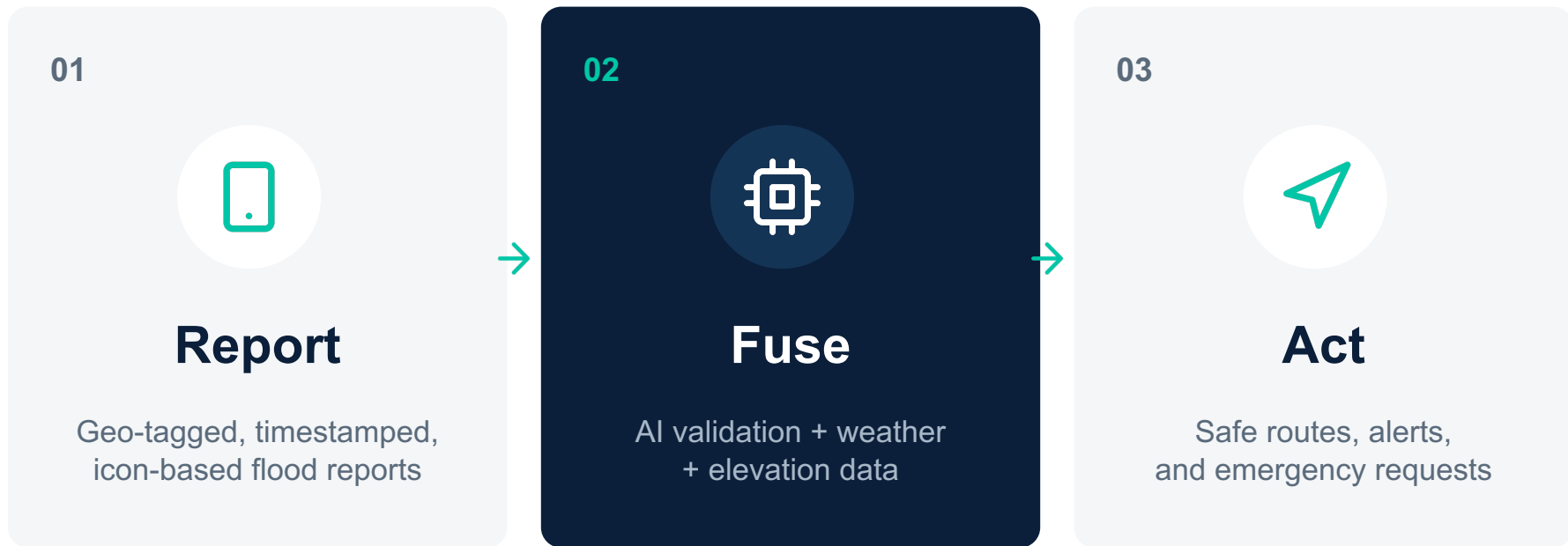
A safe road upstream can be deadly downstream



## **No shared operational picture**

Coordination is slow and reactive

# HydroMesh turns community observations into actionable flood intelligence.



# One event. Two views. Faster coordination.



## Resident mobile app

- 1 Report flooding
- 2 View risk map
- 3 Get safe route
- 4 Trigger SOS

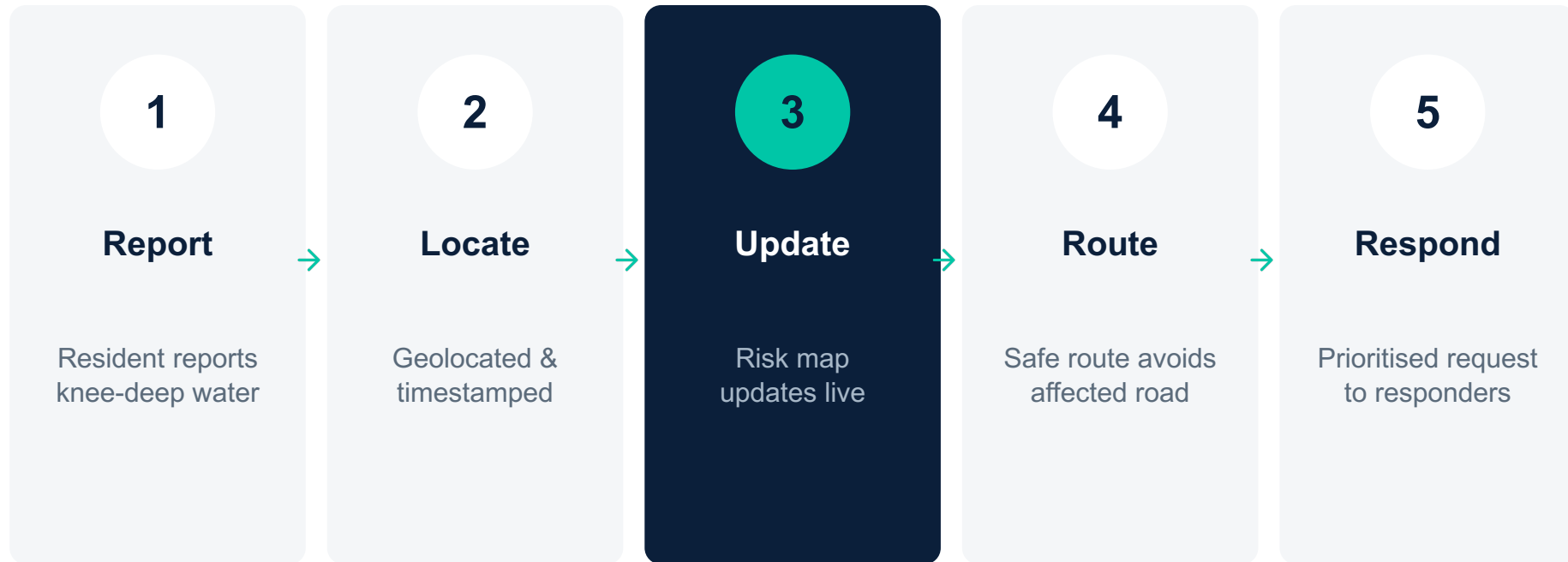


## Responder dashboard

- 1 See reports on map
- 2 Track hazard zones
- 3 Prioritise requests
- 4 Coordinate response

# From a street report to an operational response.

Demo: Simulated flood conditions



# A modular architecture connects people, data, and decisions.

Community  
reports

Weather  
data

Map &  
elevation

Node.js API · Express · Socket.io

PostgreSQL + PostGIS

Risk processing · Validation · OSRM routing

Flutter mobile app

Responder dashboard

Flutter · Node.js · Express · Socket.io · PostgreSQL/PostGIS · OSRM · Open-Meteo · Render

# The core prototype is complete and ready for controlled testing.

Flood reporting

Built

Backend integration

Built

Geolocation

Built

Municipal dashboard

Next phase

Live map

Built

LSTM forecasting

Research

Safe routing

Built

Sentiment / news layer

Research

SOS / help requests

Built

RL resource allocation

Future



# We have validated the system as an integrated academic prototype.



**Distinction-grade  
university project**

University of Birmingham  
SDG 11



**FII-MIT competition  
finalist**

Reached final rounds  
of international challenge



**MCN UN fellowship  
selected**

International recognition  
for climate resilience



**122 GitHub commits  
end-to-end tested**

Full-stack prototype  
ready for pilot partners

# The next layer is decision support for municipal teams.

Planned R&D — not yet deployed



## Live flood map

Reports, weather, sensors,  
affected roads



## Forecasting

LSTM short-horizon  
risk estimates



## Public signal monitoring

News & sentiment with  
confidence levels



## Response planning

Incidents, resources,  
shelters, routes

# We are looking for one city or institutional partner to validate HydroMesh in practice.

**Weeks 1–2**

**City data &  
stakeholder setup**

**Weeks 3–4**

**Community onboarding  
& configuration**

**Weeks 5–6**

**Controlled event  
testing**

**Weeks 7–8**

**Evaluation &  
impact report**

## **Pilot KPIs**

Report → map update time

Verified reports

Route generation time

SOS acknowledgement

Active users

False-alert rate

Coordination time

# Help us move from validated prototype to real-world pilot.

01

## Pilot partner

Municipal agency, NGO,  
or university

02

## Technical support

Data, validation, and  
deployment expertise

03

## Funding

Testing, infrastructure,  
accessibility & models

*HydroMesh makes flood intelligence local, actionable, and accessible.*

[sxm2114@student.bham.ac.uk](mailto:sxm2114@student.bham.ac.uk) · [github.com/theSaksham02/Hydromesh](https://github.com/theSaksham02/Hydromesh) · [hydromesh.vercel.app](https://hydromesh.vercel.app)



# Output Outlaws — seven disciplines, one mission.

APPENDIX

SM

**Saksham Mishra**

Project Lead · Backend · Map UI

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**Shaazia Raziq**

DB Architect · PostGIS

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**Moustafa Ameen**

UI/UX · Flood Report

FA

**Fartun Araye**

Accessibility · TTS · Contrast

DS

**Dhwanit Soni**

Safety Systems · SOS

YG

**Yaman Gulcan**

Integrations · Weather · Risk

AK

**Adham Khashan**

Documentation · Compliance

# Training data and AI models for future forecasting.

APPENDIX · Research phase — planned integration

## Groundsource

2.6M flood events

## Open-Meteo

Historical weather

## Community reports

Growing live dataset

LSTM encoder–decoder

## 24-hour flood risk forecast

Planned research integration — not in current prototype

# Thank you.

Questions?

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